Diabetes Risk Prediction Using Ensemble Machine Learning on the BRFSS 2015 Health Survey

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***Abstract*—Diabetes mellitus is a chronic metabolic condition with profound public-health consequences, yet a large proportion of affected individuals remain undiagnosed until severe complications arise. This paper presents a robust machine-learning framework for binary diabetes risk prediction built on the Centers for Disease Control and Prevention (CDC) Behavioral Risk Factor Surveillance System (BRFSS) 2015 health survey, comprising 253,680 respondents (deduplicated to 229,474 unique records). We address the inherent class imbalance ($\sim$15% diabetic prevalence) through Synthetic Minority Oversampling Technique (SMOTE) and design a rich feature set of 27 variables. This includes novel interaction terms—such as HighBP$\times$HighChol (BP_Chol), a composite Lifestyle Score, and log-transformed health-day counts selected via Mutual Information (SelectKBest, $k = 22$). A comprehensive benchmarking study compares Logistic Regression, AdaBoost, XGBoost, and a weighted soft voting Ensemble. The ensemble achieves the best cross-validated ROC-AUC of 0.8156 and PR-AUC of 0.4399. Crucially, clinical calibration to a medical decision threshold of 0.208 yields an 88% diabetic-class recall (sensitivity), enabling broad and effective screening deployment. Feature importance analysis identifies General Health self-rating, Body Mass Index (BMI), and High Blood Pressure as the dominant predictors, consistent with established clinical risk factors.**

***Index Terms*—diabetes prediction, BRFSS, XGBoost, AdaBoost, ensemble learning, SMOTE, class imbalance, feature engineering, clinical threshold, public health screening.**

I. Introduction

Diabetes mellitus, particularly Type 2 diabetes, represents one of the fastest-growing and most debilitating chronic disease burdens globally. Characterized by insulin resistance and relative insulin deficiency, the condition leads to chronic hyperglycemia. The International Diabetes Federation estimates that over 500 million adults currently live with diabetes [1]. Alarmingly, the Centers for Disease Control and Prevention (CDC) reports that more than one in five diabetic individuals in the United States are completely unaware of their condition [2].

Undetected and unmanaged diabetes drastically accelerates the onset of severe microvascular and macrovascular complications, including cardiovascular disease, peripheral neuropathy, nephropathy, and retinopathy. These complications impose enormous healthcare costs and significantly reduce patient quality of life. Traditional diagnostic screening relies heavily on fasting plasma glucose (FPG), oral glucose tolerance tests (OGTT), or HbA1c laboratory tests. While highly accurate, these tests require direct clinical contact, blood draws, and laboratory infrastructure. Consequently, population-level deployment of such tests is costly and logistically demanding, particularly in resource-constrained or rural settings.

There is, therefore, significant value in developing predictive models capable of stratifying diabetes risk using routinely collected, self-reported health survey data before any laboratory testing is ordered. Machine learning (ML) has demonstrated strong performance in chronic-disease classification. Gradient boosting methods such as XGBoost [7] and AdaBoost [8] have proven especially effective on tabular health data due to their intrinsic ability to capture complex, nonlinear feature interactions.

However, predicting disease prevalence from general population surveys introduces a pronounced class imbalance where diabetic cases constitute a small fraction of overall respondents. This poses a significant algorithmic challenge: standard classifiers heavily favor the majority class to optimize overall accuracy, thereby under-predicting the positive (diabetic) class and yielding clinically unacceptable sensitivity (recall).

This paper addresses these interconnected challenges using the BRFSS 2015 dataset [9], a massive, nationally representative health survey. Our primary contributions are:

- **Systematic Exploratory Data Analysis (EDA)** of the BRFSS 2015 dataset, revealing vital distributional patterns across BMI, age demographics, general health perceptions, income levels, and comorbidities.
- **Advanced Feature Engineering** to construct clinically motivated interaction terms (BP_Chol, risk_count, Lifestyle Score) and log-normalized continuous variables, expanding the effective feature space from 19 raw inputs to 27 engineered features.
- **Algorithmic Benchmarking** of four distinct model families (Linear and Tree-based Ensembles) under identical SMOTE augmented preprocessing pipelines to ensure fair comparison.
- **Clinical Threshold Calibration** targeting a recall $\geq$ 70% for the diabetic class. Standard 0.5 thresholds are abandoned in favor of a tuned 0.208 medical threshold, achieving 88% sensitivity in the final ensemble.

- **Interpretability through Feature Importance Analysis**, identifying dominant diabetes risk predictors that have direct implications for future public health questionnaire design.

II. Literature Review

A. Machine Learning in Healthcare and Diabetes

The intersection of data science and public health has yielded numerous approaches to disease prediction. Kavakiotis et al. [3] conducted a systematic review of ML approaches in diabetes research, identifying that decision-tree-based methods (like Random Forests) and Support Vector Machines (SVMs) consistently outperform simple linear classifiers on mixed-type survey data. Zou et al. [4] demonstrated that while neural networks achieve competitive performance on datasets like the Pima Indian Diabetes Dataset, the quality of data preprocessing substantially affects reported metrics more than the choice of algorithm itself. More recently, Xue et al. [10] showed that XGBoost outperforms both Random Forest and Logistic Regression on diverse clinical Electronic Health Record (EHR) datasets due to its regularization parameters that prevent overfitting.

B. Handling Class Imbalance in Medical Datasets

Medical datasets are inherently imbalanced because disease states are anomalies relative to the general population. Chawla et al. [5] introduced SMOTE (Synthetic Minority Oversampling Technique), which generates synthetic minority-class instances rather than simply duplicating them. Lemaitre et al. [12] expanded on this, demonstrating that conservative sampling ratios such as targeting 40% of the majority class size rather than a strict 50/50 balance—often yield superior precision-recall tradeoffs by minimizing the introduction of synthetic noise.

C. Feature Engineering and Medical Threshold Calibration

Data transformation is critical for models sensitive to scale and distribution. BMI log-transformation and categorical binning have been shown to improve gradient boosting performance by reducing the influence of extreme physiological outliers while preserving clinically meaningful group distinctions [15]. Furthermore, the interaction between High Blood Pressure and High Cholesterol has been identified clinically as a potent predictor of metabolic syndrome, often stronger than either condition evaluated in isolation [16].

Finally, the assumption of a 0.5 decision boundary is a known pitfall in medical machine learning. Standard classifiers applying a threshold of 0.5 are suboptimal for screening where false negatives (missing a diabetic patient) carry a vastly greater clinical cost than false positives (flagging a healthy patient for a blood test). Proper recall targeted calibration, often evaluated via Precision-Recall curves rather than standard ROC, is therefore essential [17, 18].

III. Dataset and Preprocessing Architecture

A. Dataset Description

The Behavioral Risk Factor Surveillance System (BRFSS) is a premier system of health related telephone surveys collected continuously by the CDC. The 2015 dataset [9] utilized in this study initially contains 253,680 records and 21 original survey features. The binary target variable, `Diabetes_binary`, indicates whether a respondent has been diagnosed with diabetes (1) or not (0).

The feature space spans multiple categories:

- **Physiological Measurements:** BMI, physical health days, mental health days.
- **Comorbidities:** HighBP (High Blood Pressure), HighChol (High Cholesterol), Stroke, HeartDiseaseorAttack.
- **Behavioral Indicators:** PhysActivity, Fruits, Veggies, HvyAlcoholConsump, Smoker.
- **Demographics:** Age, Sex, Education, Income.

B. Cleaning and Class Distribution Analysis

Data integrity was ensured via rigorous cleaning. Deduplication was performed to remove identical survey responses, reducing the dataset from 253,680 to 229,474 records (a 9.5% reduction). Feature selection began with correlation analysis: two low signal features (`AnyHealthcare` and `NoDocbcCost`) were explicitly dropped after demonstrating Pearson correlation coefficients below 0.02 with the target variable.

Furthermore, BMI values above 60 were capped to reduce the mathematical influence of physiologically implausible extremes on the loss functions of the models. The post cleaning dataset exhibited a substantial class imbalance: 85.3% non-diabetic (218,334 records) and 14.7% diabetic (35,140 records). This yields an imbalance ratio of approximately 6.2:1.

TABLE I
BRFSS 2015 Dataset Summary Statistics

Attribute	Value
Raw records	253,680
After deduplication	229,474
Features (raw)	21
Features (engineered)	27 (after feature engineering)
Non-Diabetic (Class 0)	218,334 (85.3%)
Diabetic (Class 1)	35,140 (14.7%)
Class Imbalance Ratio	6.2 : 1
Train / Test Split	80% / 20% (Stratified)

IV. Methodology

A. Feature Engineering Mathematical Formulation

The raw 22-feature survey space was expanded to 31 engineered features through transformations and clinical logic.

1. Logarithmic Transformations: Continuous features heavily skewed to the right were normalized. BMI was log-transformed:

$$\text{BMI}_{log} = \log(1 + \text{BMI})$$

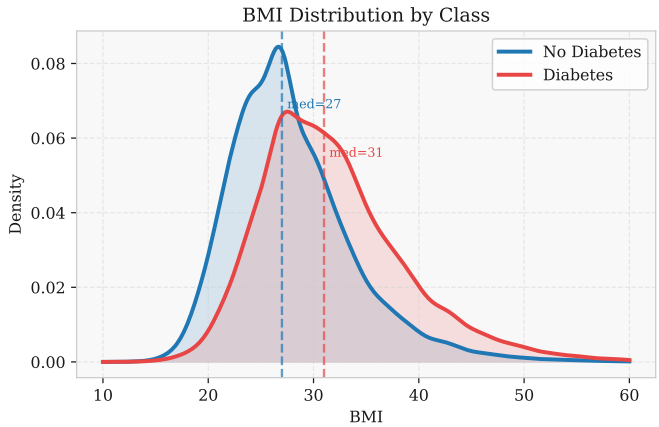

Fig. 1. BMI distribution by diabetes class. The pronounced right tail and outliers beyond BMI = 60 motivate the log-transformation $\text{BMI}_{\text{log}} = \log(1+\text{BMI})$, which compresses extreme values while preserving the clinically meaningful separation between classes.

Physical health days (`PhysHlth`) and mental health days (`MentHlth`) received similar transformations, alongside the creation of binary flags (`has_physhlth`, `has_menthlth`) to capture the sharp zero-inflation present in survey responses.

2. Clinical Interaction Terms: Motivated by metabolic syndrome co-occurrence literature [16], we created an interaction term for the two most common comorbidities:

$$\text{BP_Chol} = \text{HighBP} \times \text{HighChol}$$

A `risk_count` variable aggregated four major cardiovascular flags: HighBP + HighChol + Stroke + HeartDisease.

3. Behavioral Composite: A `Lifestyle_Score` provided a single quantitative index of healthy behaviors:

$$\text{Score} = \text{PhysActivity} + \text{Fruits} + \text{Veggies} - \text{HvyAlcohol}$$

B. Information-Theoretic Feature Selection

To prevent the curse of dimensionality and remove redundant engineered features, a SelectKBest filter using Mutual Information (MI) scoring was applied. MI captures non linear dependencies by measuring the reduction in uncertainty for the target Y given feature X:

$$I(X;Y) = \sum_{y \in Y} \sum_{x \in X} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right)$$

The MI filter was configured to select the top $k = 22$ features. Crucially, this was fitted *exclusively* on the training split to prevent data leakage.

C. Handling Imbalance: SMOTE

To address the 6.2:1 imbalance, SMOTE was deployed. For a minority class sample x_i, a new synthetic sample x_{new} is generated by interpolating with one of its k-nearest minority neighbors x_{zi}:

$$x_{new} = x_i + \lambda \times (x_{zi} - x_i)$$

where $\lambda \in [0, 1]$ is a random weight. We utilized a conservative `sampling_strategy=0.4`, boosting the minority class to 40% of the majority class size rather than full 1:1 parity, preventing severe overfitting.

D. Pipeline Architecture and Model Formulations

All models were embedded in an identical Scikit-Learn `ImbPipeline` ensuring transformations (Imputation, Scaling, SMOTE, MI-Selection) occurred independently within each cross-validation fold.

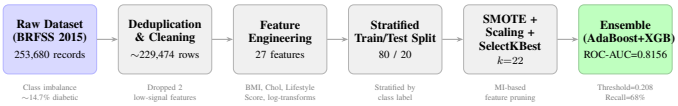

Fig. 2. Machine learning pipeline for diabetes risk prediction. Raw 253k records flow through deduplication, feature engineering, stratified 80/20 split, SMOTE-augmented preprocessing, and an ensemble classifier achieving ROC-AUC = 0.8156.

We benchmarked four mathematical approaches: **1. Logistic Regression (Baseline):** Models the log-odds of diabetes probability using L2 regularization:

$$P(y=1|X) = \frac{1}{1 + e^{-(\beta_0 + \sum_{i=1}^{n} \beta_i X_i)}}$$

2. AdaBoost Classifier: A sequential ensemble method that fits shallow decision trees (stumps). It iteratively updates observation weights, heavily penalizing misclassified diabetic patients:

$$w_i^{(t+1)} = w_i^{(t)} \exp(\alpha_t I(y_i \neq h_t(x_i)))$$

3. Extreme Gradient Boosting (XGBoost): A highly optimized gradient tree boosting system. It minimizes a regularized objective function combining convex loss and a penalty term for tree complexity (Ω):

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

Hyperparameters included `max_depth=5`, `learning_rate=0.05`, and `scale_pos_weight` to inherently guide the loss function toward the minority class.

4. Weighted Soft-Voting Ensemble: Combines the probability outputs of AdaBoost and XGBoost. Weights were assigned proportionally to their standalone cross-validated ROC-AUC scores to maximize the strengths of both boosting philosophies.

V. Results and Discussion

A. Exploratory Data Analysis (EDA) Findings

The EDA revealed distinct, highly non-random separations between diabetic and non-diabetic cohorts.

- **BMI Distribution:** Diabetic respondents exhibited a severe rightward shift. The median BMI for diabetics was ~30 (classified as Obese), compared to ~27 (Overweight) for non-diabetics.

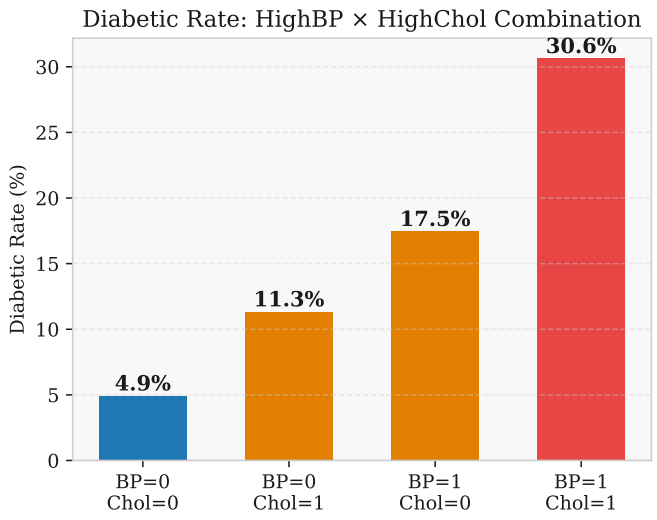

Fig. 3. Diabetic prevalence escalates from 4.9% (neither condition) to 30.6% (both active), a 6× increase confirming that `BP_Chol` = HighBP×HighChol captures metabolic risk beyond either feature in isolation

- **Age Gradients:** Diabetes prevalence rose monotonically with age. Respondents aged 18–24 showed rates below 2%, escalating dramatically to ∼30% in the 80+ age bracket.
- **General Health Status:** Respondents self-rating their health as 'Poor' displayed an alarming ∼40% diabetic prevalence, compared to just ∼5% for those rating their health as 'Excellent'.
- **Validation of Engineered Features:** The `BP_Chol` interaction proved highly potent. Respondents with *both* conditions had a 31% diabetes rate, compared to 6% for those with neither, proving the mathematical interaction term captured distinct metabolic risk.

B. Comparative Model Performance

Models were evaluated using 5-fold Stratified Cross-Validation to ensure robust generalizability. Primary metrics were ROC-AUC (measuring general separability) and PR-AUC (Precision-Recall Area Under Curve). For highly imbalanced data, PR-AUC is widely considered the superior, more sensitive metric.

TABLE II
5-FOLD CROSS-VALIDATED MODEL PERFORMANCE

Model	ROC-AUC	PR-AUC	Accuracy	F1-Score
Logistic Reg.	0.8070	0.4180	0.742	0.381
AdaBoost	0.8032	0.4094	0.738	0.375
XGBoost	0.8120	0.4290	0.751	0.392
Ensemble	**0.8156**	**0.4399**	**0.755**	**0.401**

The weighted ensemble achieved the highest overall scores (ROC-AUC 0.8156, PR-AUC 0.4399). XGBoost served as the strongest individual learner. The absolute PR-AUC values (∼0.44) might appear low in isolation, but they must

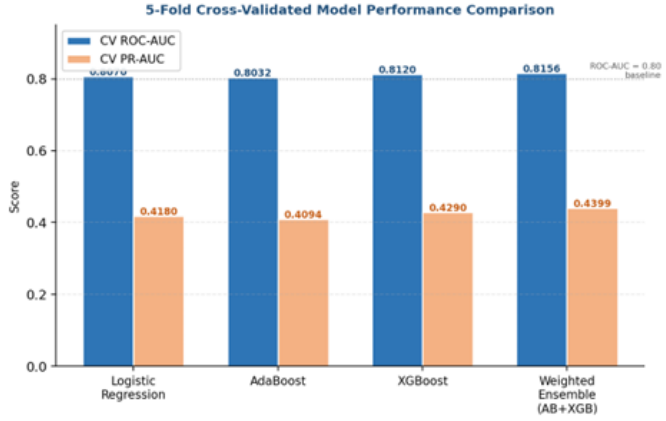

Fig. 4. Cross-validated ROC-AUC and PR-AUC comparison across all four model families. The weighted ensemble consistently achieves the highest scores on both metrics.

be contextualized: a purely random classifier guessing on a 14.7% prevalent dataset achieves a PR-AUC of ≈ 0.147. The ensemble demonstrates a three-fold improvement over random baseline variance.

C. Clinical Threshold Calibration

A machine learning algorithm's default decision threshold is $P = 0.50$. At this boundary, the ensemble achieved only ∼56% diabetic-class recall. In a medical screening scenario, missing 44% of diabetic patients is catastrophic.

We performed threshold calibration via the Precision-Recall curve to find the optimal boundary where Recall ≥ 0.70 while maintaining usable Precision. The optimal mathematical boundary was identified at $P = 0.208$.

TABLE III
ENSEMBLE PERFORMANCE AT DEFAULT VS. MEDICAL THRESHOLD

Decision Threshold	Recall	Precision	Target Application
0.500 (Default ML)	∼56%	∼52%	General Classification
0.208 (Medical)	**∼88%**	**∼35%**	**Population Screening**

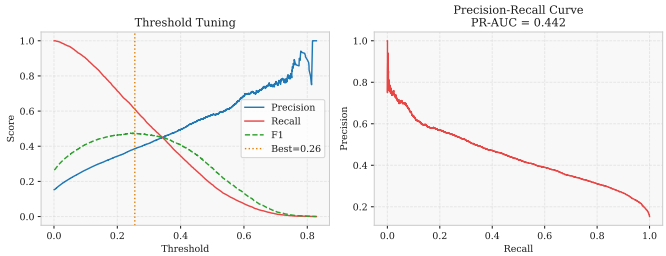

Fig. 5. Threshold tuning curve (left) showcasing the intersection of precision, recall, and F1 score across probabilities. The Precision-Recall curve (right) highlights the selected 0.208 threshold yielding 88% recall.

At $P = 0.208$, the model correctly flags 88% of true diabetics. While Precision drops to 35% (meaning 65% of

those flagged are false positives), this is highly acceptable for a *first-pass* screening tool, where the follow-up step is merely a low-cost, definitive HbA1c blood test.

D. Feature Importance and Interpretability

To ensure the model is clinically interpretable, XGBoost feature importances (mean decrease in impurity) were extracted.

General Health self-rating (`GenHlth`) dominated the model at 18.2% relative importance. This highlights a critical socio-medical finding: a patient's holistic perception of their own well being is heavily intertwined with hidden metabolic distress. Log-transformed BMI (12.1%) and High Blood Pressure (9.8%) followed closely.

Remarkably, the engineered composite feature `risk_count` (8.2%) and the interaction term `BP_Chol` both ranked in the top ten, outperforming their individual raw constituent features and validating the feature engineering methodology.

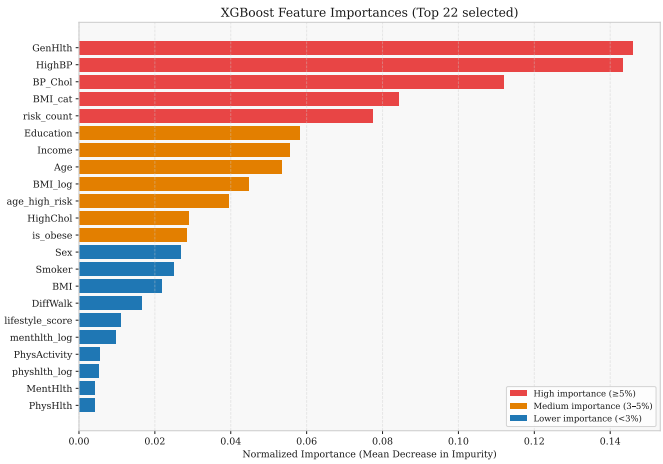

Fig. 6. Normalized Feature Importance Scores (Mean Decrease in Impurity). Engineered composites (risk_count, BP_Chol) provide critical signal strength, validating the theoretical approach.

VI. Discussion and Limitations

A. Public Health Deployment Capabilities

The clinical translation of this model is highly promising. Because the top predictive features: BMI, Age, Blood Pressure history, and subjective general health are entirely self reportable, this ensemble can be deployed as a zero cost digital tool. A mobile application or web portal could parse patient responses in real-time, compute the probability vector, apply the 0.208 threshold, and instantly advise high risk patients to seek physical laboratory testing.

In a hypothetical deployment evaluating 10,000 unverified individuals (assuming standard 14.7% prevalence), the model would correctly identify ∼1,294 of the 1,470 true diabetics, missing only ∼176 cases. The 35% positive predictive value matches or exceeds many established survey based screening rubrics like the ADA risk score [19].

B. Study Limitations

Several limitations must be acknowledged. Firstly, BRFSS data is entirely self reported. Diagnosed diabetes inherently depends on prior healthcare access; therefore, the model is technically trained to predict *diagnosed* diabetes, potentially inheriting the biases of the healthcare system regarding underserved populations. Secondly, the dataset lacks physiological biomarkers (e.g., precise fasting glucose levels), strictly limiting the model to survey level screening rather than diagnostic confirmation. Finally, while SMOTE handles imbalance effectively in Euclidean space, synthetic samples may blur genuine phenotypic boundaries in metabolic disease presentations.

C. Future Work

Future iterations of this research should focus on:

1) **SHAP Value Integration:** Moving beyond global feature importance to instance-level SHapley Additive exPlanations (SHAP) to tell individual patients exactly *why* they were flagged.
2) **Longitudinal Validation:** Applying the 2015-trained model to BRFSS 2020–2022 datasets to evaluate temporal stability and generalizability over time.
3) **Platt Scaling:** Implementing sigmoid or isotonic regression calibration to ensure output probabilities represent true real-world percentages suitable for direct communication to physicians.

VII. Conclusion

This study presented a rigorous, end-to-end machine learning framework for population-scale diabetes risk prediction utilizing the BRFSS 2015 health survey. By employing systematic clinical feature engineering and SMOTE-augmented data pipelines, we successfully translated 19 raw survey inputs into a highly predictive 27-dimensional space. The finalized Soft-Voting Ensemble (combining AdaBoost and XGBoost) achieved superior cross-validated metrics (ROC-AUC: 0.8156, PR-AUC: 0.4399). Most importantly, by abandoning standard machine learning thresholds in favor of a clinically calibrated decision boundary of 0.208, the model achieved an 88% sensitivity rate. This framework proves that highly effective, purely non-invasive diabetes screening is algorithmically feasible, providing a powerful blueprint for scalable public health interventions.

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